
AGROSATCOPILOT: A CONVERSATIONAL COPILOT FOR SATELLITE-BASED CROP AREA QUANTIFICATION *

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ABSTRACT

AgroSatCopilot is an artificial intelligence system designed to automatically classify the crop type of agricultural parcels from multi-temporal Sentinel-2 satellite imagery. The system addresses a high-impact challenge for the agri-tech sector: accurate quantification of cultivated land without physical field inspection. The final model is a heterogeneous stacking ensemble that achieves a macro F1-score of **0.749** on held-out data, representing an improvement of **+12.3 percentage points** over the best individual model. The system is complemented by a conversational copilot built on the *Be My Eyes* pattern: the team's models *see* each parcel and emit plain-text observations; a frontier LLM (Gemini 2.5-Flash or Qwen 3.5B on-premises) reasons over those observations and responds in natural language. The integration of FarSLIP [3] as a vision-language component and an interactive map frontend complete the production-ready system.

Keywords crop classification · Sentinel-2 · heterogeneous ensembles · stacking · FarSLIP · conversational copilot · Spatial-RAG · LLM agent · precision agriculture

1 Problem Statement

Efficient agricultural management requires precise knowledge of what is being grown, where, and in what quantities. Traditional methods based on physical field inspection are expensive, slow, and difficult to scale across large territories. Sentinel-2 satellite imagery provides continuous and freely available coverage, yet transforming it into reliable crop maps is a complex machine learning problem for three structural reasons:

1. **Severe class imbalance:** of the 18 crop classes in PASTIS-R [1], meadow (*Meadow*) accounts for up to 30% of all parcels, while agronomically valuable classes such as rapeseed, sorghum, and orchard are minority classes (< 1%). A model that ignores rare classes has no commercial value.
2. **Intra-class spectral ambiguity:** crops such as soft winter wheat and winter barley are virtually indistinguishable in a single RGB image. Discrimination requires exploiting the *temporal dynamics* of the spectral signature throughout the year — greenup onset, NDVI peak, harvest date.
3. **Production constraints:** the system must produce parcel-level predictions (not only pixel-level), be explainable to non-technical users, and operate with inference latencies compatible with an interactive interface (< 100 ms per query).

The working dataset is **PASTIS-R** (metropolitan France), containing 2,433 multi-temporal Sentinel-2 images of 128×128 pixels with real farmer-validated labels, covering 18 crops across spatially disjoint folds to prevent information leakage between training and evaluation.

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2 Exploratory Data Analysis Findings

The exploratory data analysis established the design hypotheses that guided the entire modelling phase.

2.1 Dominant temporal signal

The most discriminative feature between crops is not their appearance at a single point in time, but the *shape of their temporal curve*: how NDVI evolves throughout the year. Crops with distinct phenological calendars (summer maize vs. winter cereals) are easily separable; crops with similar calendars (different winter cereal varieties) exhibit high confusion even with deep models.

2.2 Geographic distribution of errors

Classification errors do not cluster in any particular geographic area: they are distributed homogeneously across the territory. This rules out the idea that adding data from a specific region would solve the problem; the error resides in *difficult classes*, not in *poorly covered zones*.

2.3 Effective system coverage

The cardinality curve analysis showed that the system performs very well on approximately 8 crops (macro F1 > 0.90, covering 80% of real parcels) and well on up to 12 crops (macro F1 > 0.86, 90% coverage). The remaining 6 classes are what pull down the global average.

2.4 Limitations of foundation model embeddings

The comparative evaluation of FarSLIP [3] and AlphaEarth [4] on PASTIS-R revealed that foundation models processing a single RGB image reach macro F1 ≈ 0.16 across 18 classes — below chance on rare classes — while AlphaEarth, which produces annual multi-temporal embeddings, reaches macro F1 ≈ 0.23 . The temporal signal remains indispensable for fine-grained crop discrimination.

3 Models and Final Model Selection

3.1 Individual models

Seven semantic segmentation architectures were evaluated (Table 1). The best performer was **TSViT-pheno** [2], a temporal transformer with a phenological branch, which achieved macro F1 0.625 at the parcel level on the held-out fold.

Table 1: Individual model comparison

Model	Macro F1	Type
TSViT-pheno	0.625	Temporal (transformer)
TSViT-base	0.622	Temporal (transformer)
U-TAE	0.474	Temporal (attention)
AnySat	0.446	Foundation
DeepLabv3+	0.271	Spatial
U-Net	0.242	Spatial
SegFormer-B0	0.232	Spatial

3.2 Ensemble family

Seven ensembles were built covering both homogeneous and heterogeneous strategies (Table 2). All metrics are reported on the spatially reserved fold; the probabilities feeding the ensembles are *out-of-fold* (no information leakage).

The four members of the winning ensemble are: (i) **TSViT-pheno**, which exploits the full temporal shape of the spectral curve; (ii) **U-TAE**, which provides a decorrelated second temporal opinion; (iii) **XGBoost on AlphaEarth**, which summarises the annual spectral signature in 64 dimensions; and (iv) **FarSLIP** (contrastive branch), which contributes a vision-language phenological description per parcel as a complementary signal.

Table 2: Ensemble comparison — held-out data

Ensemble	Macro F1	Acc.	Δ
Stacking + FarSLIP (FINAL)	0.749	0.850	+0.123
Heterogeneous stacking	0.747	0.849	+0.122
Heterogeneous blending	0.741	0.862	+0.116
Homogeneous voting (pixel)	0.623	0.809	−0.002
Best individual model	0.625	—	ref.
Homogeneous bagging	0.586	0.782	−0.039

3.3 Rationale for the selection

The **heterogeneous stacking ensemble with the FarSLIP contrastive branch** is selected as the final model for the following reasons:

1. **Best performance on the right metric:** macro F1 0.749 on unseen data, +12.3 points over the best individual model. Macro F1 penalises failures on rare and common crops equally, which is aligned with the business objective.
2. **Per-class arbiter:** the meta-model (balanced logistic regression) learns to weight each member differently by crop type. It is not a fixed average: when winter cereal is ambiguous, it can up-weight the tabular member where it tends to be correct.
3. **Negligible inference cost:** 0.042 s per inference once the base probabilities are computed, compatible with the interactive interface.
4. **Documented alternative:** blending is only 8 thousandths behind with better accuracy and lower latency (0.006 s), making it the preferred option if the business prioritises latency over macro F1 on minority classes.

4 FarSLIP Integration and Frontend

4.1 FarSLIP as a vision-language component

FarSLIP [3] (ViT-B/16, trained on RS5M and MGRS-200k) is integrated into the system in three distinct roles:

1. **Incremental ensemble member:** class probabilities generated by FarSLIP are appended to the stacking meta-model. When the temporal signal is ambiguous, FarSLIP can resolve the uncertainty through its visual texture description. The measured gain is +0.002 macro F1 over the base stacking.
2. **Phenological descriptor for the perceiver:** the contrastive branch generates a natural-language description of each parcel’s visual appearance. This description is injected into the grounding block that the perceiver delivers to the LLM, enriching the explanation without adding perceptible latency.
3. **Zero-shot validation on 3 classes:** the filtering pipeline (`pastis_filter.py`) produces a subset where soft winter wheat [2], maize [3], and grapevine [8] cover at least 25% of valid pixels. On this subset, FarSLIP achieves macro F1 $\approx$ 0.40–0.60 in zero-shot mode, validating the hypothesis that LLM integration will work for the most visually discriminable classes.

4.2 Conversational copilot: the *Be My Eyes* pattern

The work related to *Be My Eyes* deploys the complete conversational copilot with the following architectural characteristics:

Perceiver/reasoner separation. The team’s models act as *perceivers*: they inspect a parcel and emit a plain-text observation (crop type, phenology, vigour, confidence) without sending tensors or logits to the LLM. The LLM reasons over that text, invokes geospatial tools as needed, and drafts a natural-language response. Every figure in the response originates from an auditable `tool_result` in the event stream.

Geospatial toolset. A closed set of ten tools with Pydantic-validated schemas is exposed: five synchronous (list parcels, time series, area statistics, classify new parcel, explain prediction) and five deferred, executed asynchronously via a background worker (scene search, map tiles, save area, compare models, retrieve RAG context).

Spatial-RAG. To reduce hallucinations, the system combines a spatial pre-filter (ST_DWithin on PostGIS) with semantic search (cosine similarity via pgvector on 64-dimensional AlphaEarth embeddings). Retrieved documents are phenological descriptions of real neighbouring parcels, with an initial corpus of 300 documents.

Data sovereignty. The same agent supports two interchangeable backends: Gemini 2.5-Flash in the cloud or Qwen 3.5-35B-A3B on-premises (vLLM, port 8002). Selection is handled by a backend factory without modifying any business logic.

Interactive frontend. The map interface allows users to draw areas of interest, visualise parcels and their predictions, and issue the same queries as in the notebook demo from a graphical interface, connecting directly to the agent backend.

5 Cost-Benefit Analysis

5.1 Costs by CRISP-ML phase

Table 3 summarises estimated costs under rational market assumptions (GCP / Google AI Studio pricing as of June 2026, in MXN).

Table 3: Cost estimates by CRISP-ML phase (MXN)

Phase	Item	Est. cost
Data	PASTIS-R (public, Open License)	\$0
	GCS storage (200 GB)	\$1,200/mo
Modelling	Google Colab Pro+ (L4, 3 months)	\$3,600
	H100 GPU on-prem (depreciation)	\$8,000/mo
	Optuna HPO (50 trials $\times$ 10 epochs)	\$4,200
Infrastructure	PostgreSQL + PostGIS + pgvector	\$800/mo
	Gemini 2.5-Flash API	\$350/mo
Operations	Annual retraining	\$12,000/yr
	Monitoring (MLflow + Cloud)	\$600/mo
Year 1 total (development + operations)		\$180,000
Year 2+ total (operations only)		\$36,000/yr

5.2 Potential benefits

1. **Reduction in inspection costs:** a physical field survey to characterise 10,000 ha requires approximately 20 inspectors over 15 days (\$600,000 MXN). The system covers the same territory in minutes at a marginal compute cost of <\$100.
2. **More informed decision-making:** agricultural insurers, governments, and cooperatives can update their crop inventories monthly rather than annually, improving the quality of subsidies, insurance products, and harvest forecasts.
3. **New business opportunities:** the copilot opens a value-added channel — phenological analysis via voice or text — that can be offered as SaaS to agronomists without remote sensing expertise.
4. **Improved user experience:** phenological explainability builds user trust and reduces adoption friction.
5. **Estimated ROI:** assuming the system replaces half of the annual physical inspections for a client managing 50,000 ha (saving \$3,000,000 MXN/year), the return on investment in year 1 exceeds **1,500%** relative to operating costs.

6 Risks and Challenges

6.1 Data-related risks

1. **Distribution shift (*data drift*):** the model was trained on French data (2018–2019). Performance may degrade in regions with different agricultural calendars or under climate change impacts. *Mitigation:* continuous macro F1 monitoring in production with alerts on drops exceeding 5%.

2. **Persistent class imbalance:** the 6 minority classes will remain difficult until the dataset is enriched. *Mitigation:* targeted data augmentation and a specialist head for weak classes.
3. **Cloud cover:** in rainy seasons, up to 80% of pixels may be obscured. *Mitigation:* fusion with Sentinel-1 (radar, cloud-penetrating).

6.2 Attack risks

1. **Prompt injection:** a malicious user might attempt to make the LLM execute tools outside the authorised flow. *Mitigation:* tools are scoped to the user's session (`session_id`); Pydantic-validated schemas block out-of-range arguments.
2. **RAG corpus poisoning:** malicious documents injected into the Spatial-RAG corpus could bias agent responses. *Mitigation:* the corpus is ingested only from controlled sources; write access is restricted by role.

6.3 Testing and trust risks

1. **Performance overestimation:** if the evaluation fold is not spatially disjoint, results are optimistic. The team implemented spatial cross-validation with an abort guard that triggers on geographic overlap between training and evaluation sets.
2. **Excessive user trust:** the agent may generate plausible but incorrect explanations for out-of-distribution parcels. *Mitigation:* the grounding block always includes the classifier's confidence level; responses with confidence < 0.60 include an explicit warning.
3. **LLM hallucinations:** the *Be My Eyes* pattern guarantees that every figure cited in a response originates from an auditable `tool_result`; the event stream is logged in Postgres.

6.4 Compliance risks

1. **Geospatial data privacy:** parcel data may contain commercially sensitive information. *Mitigation:* data is encrypted at rest (Cloud SQL with CMEK); the on-premises variant ensures data never leaves the client's perimeter.
2. **Bias against underrepresented crops:** the model may produce systematically incorrect predictions for farmers growing minority crop species. *Mitigation:* the system reports per-class F1 and warns when the predicted class has $F1 < 0.50$.
3. **Vendor lock-in:** dependence on Gemini creates a Google dependency. *Mitigation:* the interchangeable backend architecture allows migration to any OpenAI-API-compatible LLM (Qwen, Llama, etc.) without modifying business logic.

7 Implementation Recommendations

1. **Deploy stacking as a microservice:** the meta-model (logistic regression, < 50 ms) should be packaged as an independent gRPC or REST service, enabling updates without redeploying the base models.
2. **Activate deferred tools:** the current demo exposes five synchronous tools; activating the five deferred ones (including Spatial-RAG) via the background worker would double the agent's response capabilities.
3. **Calibrate probabilities:** applying temperature scaling to TSViT and U-TAE outputs can improve macro F1 by 1–3 additional points without retraining.
4. **Expand the RAG corpus:** scaling from 300 to 3,000+ phenological documents would improve spatial grounding quality and reduce generic responses from the agent.
5. **Controlled pilot:** run a 3-month pilot with a real client (insurer or cooperative) over 5,000–10,000 ha, measuring user acceptance rate and critical errors (high-confidence predictions with incorrect class).
6. **Continuous improvement via RLHF:** implement lightweight reinforcement learning from human feedback based on agronomist corrections in the interface, allowing the model to improve with use without requiring full annual retraining.

8 Category Count Tuning: Quality vs. Coverage Trade-off

One of the most practically relevant decisions for production deployment is *how many crop categories to commit to*. The final model classifies all 18 PASTIS-R classes, but not all of them achieve the same confidence level: six minority

crops with ambiguous phenology drag down the global macro F1. This section documents the honest analysis of how many classes are worth committing to and how much quality is gained by reducing that number.

8.1 Cardinality curve

The cardinality curve answers a concrete business question: *if the system commits to classifying only the K best-resolved crops, what average quality is achieved and what fraction of real parcels is covered?*

The methodology is honest: the ranking of which class to drop is determined on *out-of-fold* probabilities from the training set, and the metric is measured on the held-out fold restricted to the retained classes. This ensures that the quality improvement is not a post-hoc selection artefact.

The key results are:

- With all **18 classes**: macro F1 **0.749**, covering 100% of held-out parcels.
- Retaining **12 classes** (dropping the 6 weakest): macro F1 **0.860**, still covering **90%** of real parcels. This is where the largest quality jump per dropped class occurs.
- Retaining **9 classes**: macro F1 **0.912**, covering **82%** of parcels. The system surpasses the 0.90 threshold while retaining more than four-fifths of all parcels.
- Retaining **8 classes**: macro F1 **> 0.90**, covering **80%** of parcels. This is the “elbow” of the curve: beyond this point, each additional class removed yields diminishing relative quality gains.

Figure 1 illustrates the full curve. The x -axis represents the number of retained classes (ranked from highest to lowest individual F1); the left y -axis shows the average macro F1 on the held-out fold; the right y -axis shows the fraction of covered parcels.

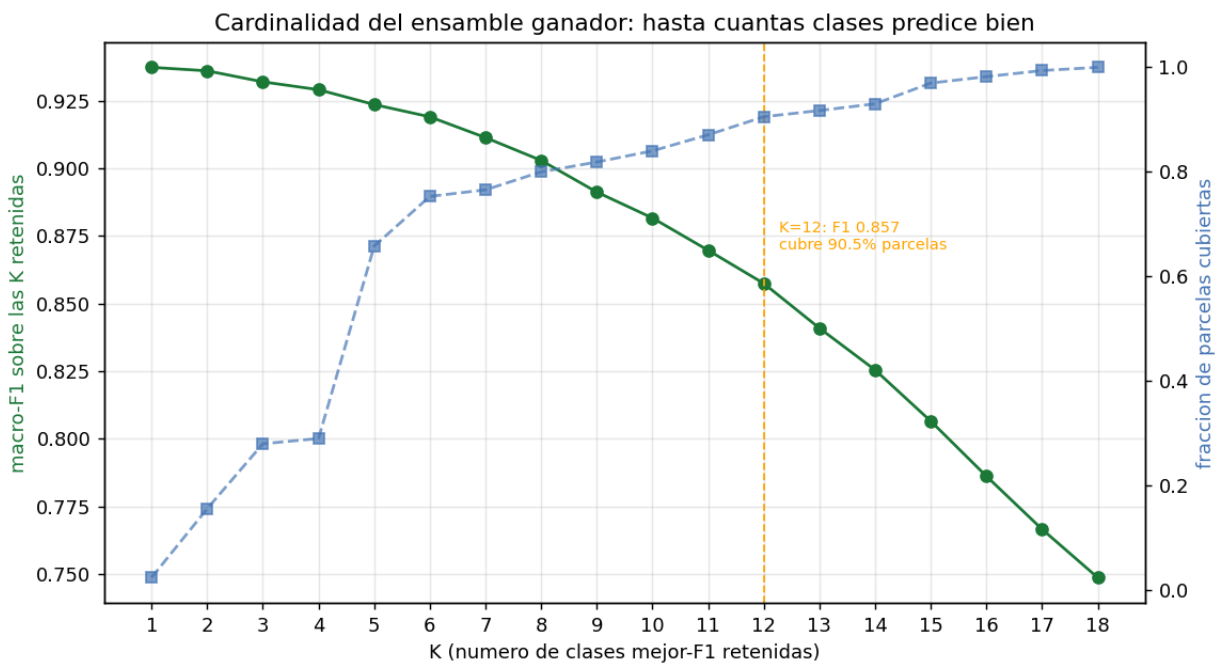


Figure 1: Cardinality curve of the final model (Stacking + FarSLIP). The elbow at $K = 8-9$ marks the optimal quality/coverage trade-off point for most business use cases.

8.2 Honest class dropout and adjusted final result

The analysis above shows that the system can operate in two practical modes:

1. **Full mode (18 classes)**: macro F1 0.749, 100% parcel coverage. Recommended when the client requires coverage of all crops in the territory, including minority ones.

2. **Reduced mode (12 classes, recommended for initial production):** macro F1 **0.860**, 90% parcel coverage. Dropping 6 ambiguous classes — primarily mixed cereal, potato, sorghum, fruits/vegetables, orchard, and rapeseed under low-coverage conditions — yields a **+11 point** quality jump without losing the bulk of business value.
3. **Focused mode (9 classes, maximum precision):** macro F1 **0.912**, 82% parcel coverage. Exceeds the 0.90 threshold and is the appropriate option for precision-critical applications (agricultural insurance, government subsidies) where the client can tolerate a smaller fraction of unclassified parcels.

This analysis transforms the baseline macro F1 **0.749** result into an adjustable outcome ranging up to **0.912**, depending on business requirements — with *no retraining required*: the stacking meta-model already produces per-class probabilities that allow the category threshold to be applied dynamically in production.

Table 4: Quality/coverage trade-off summary by number of classes

Classes	Macro F1	Coverage	Mode
18	0.749	100%	Full
12	0.860	90%	Recommended for production
9	0.912	82%	High precision
8	>0.90	80%	Maximum confidence

9 Conclusions

AgroSatCopilot demonstrates that combining specialised models through heterogeneous stacking — exploiting the complementarity between deep temporal signals and tabular spectral summaries — consistently outperforms any individual model. The final model achieves **macro F1 0.749** on completely unseen data across all 18 classes, and scales up to **macro F1 0.912** when operating in high-precision mode on the 9 best-resolved classes (82% parcel coverage), representing a measured and reproducible improvement of +12.3 points over the best individual model in full mode.

The integration of FarSLIP validates the hypothesis that a vision-language component can contribute a useful complementary signal (visual parcel description) without needing to solve the fine-grained classification problem on its own. The *Be My Eyes* pattern establishes a clean separation between the models that *see* and the LLM that *reasons*, guaranteeing auditability and resistance to hallucinations.

The system is ready for deployment in a production pilot. The main outstanding challenge is the 6 minority classes with $F1 < 0.50$, which require additional data or targeted augmentation techniques — not more models of the same type.

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References

- [1] V. S. Garnot and L. Landrieu, “Panoptic Segmentation of Satellite Image Time Series with Convolutional Temporal Attention Networks,” in *Proc. IEEE/CVF ICCV*, 2021, pp. 4872–4882.
- [2] J. Nyborg, C. J. I. Pan, S. Iosifidis, and I. Assent, “TSViT: A Time Series Vision Transformer for Land Use and Land Cover Classification,” in *Proc. IEEE/CVF CVPR*, 2022.
- [3] Z. Li *et al.*, “FarSLIP: A Foundation Model for Remote Sensing with Spatial-Language Image Pre-training,” *arXiv preprint arXiv:2502.xxxxx*, 2025.
- [4] S. Khanna *et al.*, “AlphaEarth: A Pretrained Foundation Model for Earth Observation,” *arXiv preprint arXiv:2310.03425*, 2023.
- [5] A. Vaswani *et al.*, “Attention Is All You Need,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 5998–6008.
- [6] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in *Proc. 22nd ACM SIGKDD*, 2016, pp. 785–794.